

# CS554 Midterm #2

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## I. QUESTION

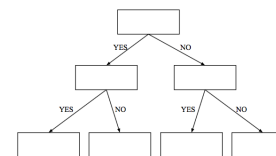
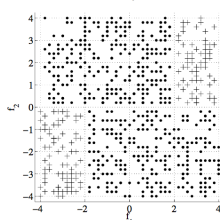
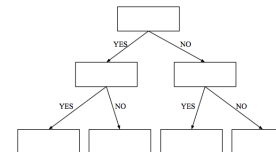
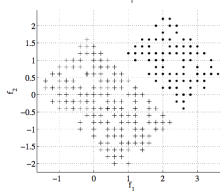
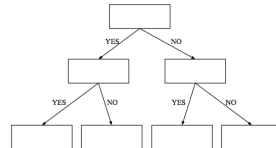
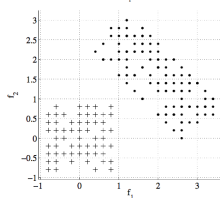
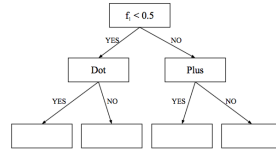
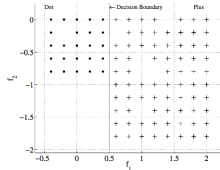
Run the linear perceptron algorithm one epoch for the following data:

| Feature 1 | Feature 2 | Class |
|-----------|-----------|-------|
| 1         | 1         | 1     |
| 1         | 0         | 1     |
| 0         | 1         | 1     |
| 0         | 0         | 0     |

Assume that the initial weights are all 0.1, and the learning rate is 0.1. Do batch update, i.e. calculate the update of the weights for all data at once.

## II. QUESTION

For the following data, fill the decision nodes with appropriate decision conditions.



## III. QUESTION

Let us say we have 4 states:  
 $S_1$ : A,  $S_2$ : C,  $S_3$ : G,  $S_4$ : T

with initial probabilities:  $\Pi = [0.3, 0.4, 0.1, 0.2]$   
 and the transition matrix is:

$$\mathbf{A} = \begin{bmatrix} 0.2 & 0.2 & 0.3 & 0.3 \\ 0.1 & 0.3 & 0.2 & 0.4 \\ 0.4 & 0.4 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.7 \end{bmatrix}$$

- Calculate the probability of the sequence ACCGTC.
- Given twelve sequences AAGCT, TATTA, GTAGT, TTATT, GAGCT, AAAAT, ACCTT, CTAA, TGATT, GGTA, TACAG, CGAAC estimate the initial probabilities for  $\Pi$  and  $\mathbf{A}$ .

## IV. QUESTION

Propose a tree induction method where each split in each node is in the form of  $a \leq x_i \leq b$ .

## V. QUESTION

Consider the alternative error function in linear perceptron for regression problem:

$$E = \frac{1}{2}(r - y)^2 + \alpha \sum_i w_i^2 \quad (1)$$

where  $r$  represents the actual output,  $y$  represents the calculated output,  $w_i$  are the weights between the input units and the output unit. Derive the gradient descent update rules for the weights.

## VI. QUESTION

If each base-learner is independently and identically distributed and correct with probability  $p > 1/2$ , what is the probability that a majority vote over  $L$  classifiers gives the correct answer? Calculate the probability for  $L = 5$  and  $p = 3/4$ .

## VII. QUESTION

Consider the three learners  $d_1$ ,  $d_2$ , and  $d_3$ ; and the posterior probabilities found for three classes for those learners.

|       | $C_1$ | $C_2$ | $C_3$ |
|-------|-------|-------|-------|
| $d_1$ | 0.2   | 0.5   | 0.3   |
| $d_2$ | 0.1   | 0.6   | 0.3   |
| $d_3$ | 0.3   | 0.3   | 0.4   |

Calculate the posterior values for three classes for sum, median, minimum, maximum and product rules.